

Adaptive Trend Filtering for ECG Denoising and Delineation

Tom Trigano, *Member, IEEE*, Shlomi Talala and David Luengo, *Member, IEEE*

Abstract—Standard recordings of electrocardiographic signals are contaminated by a large variety of noises and interferences, which impair their analysis and the further related diagnosis. In this paper, we propose a method, based on compressive sensing techniques, to remove the main noise artifacts and to locate the main features of the pulses in the electrocardiogram (ECG). The motivation is to use Trend Filtering with a varying proximal parameter, in order to sequentially capture the peaks of the ECG, which have different functional regularities. The practical implementation is based on an adaptive version of the ADMM (alternating direction method of multiplier) algorithm. We present results obtained on simulated signals and on real data illustrating the validity of this approach, showing that results in peak localization are very good in both cases and comparable to state of the art approaches.

Index Terms—ECG signal processing, compressive sensing, trend filtering, denoising, delineation.

I. INTRODUCTION

Since the development of computer-assisted signal processing, the denoising and analysis of electrocardiographic recordings remains thoroughly investigated [1]. The main reason for these developments lies in the fact that the physiological causes of the noise and interferences (artifacts), as well as their influence on the recorded signals, are well known. It is therefore important to provide accurate measurements, and to discard the perturbations that could have a bad influence on the diagnosis. Following [1], we can distinguish four common perturbations which can be observed on external ECG recordings: baseline wander (a low-frequency activity generated by numerous sources, such as body movement and respiration), as shown in Figures 1a and 1d; electrode motion (noise caused by alterations in electrode-skin impedance due to movement), as in Figures 1b and 1e; muscle (a.k.a. electromyogram (EMG)) artifacts, where the muscles' electrical activity perturbs the good recording of the heart activity, as in Figures 1c and 1f; and powerline interference, stemming from nearby electrical equipment.

Historically, discarding the noises' perturbations in recorded ECGs has been done using digital signal processing techniques. Due to their simplicity, these FIR/IIR digital filters

are still in use nowadays, and we refer to [2] for an in-depth discussion of the specifications that such digital filters should comply with for real-life apparatus. Once the signal has been denoised, the peaks are usually located on a smoother version of the signal, for example by means of an adaptive threshold, as done in [3]. However, this approach has several shortcomings, due to the compromise between amplitude and phase distortions brought by the digital filter. This issue is particularly obvious, for example, when trying to suppress the power-line interference: highly selective notch filters typically introduce ringing artifacts after the QRS complexes, which are unacceptable in practice [1]. Reducing the filter's selectivity allows us to reduce these artifacts, but comes with the price of a more contaminated signal.

Due to the previous considerations, ECG denoising and delineation remains a very active field of research, even though it has been studied for a long time. Indeed, no single method can claim to outperform all the others with respect to removing the different noises and interferences. Consequently, practitioners are forced to use specialized algorithms to remove all the perturbations in the ECG signal [7], [8]. Similarly, regarding wave segmentation and delineation, different methods provide the best performance for P, QRS and T waves, as shown in the extensive comparison undertaken in [9]. Hence, the quest for a single method that can provide a reliable performance on all types of perturbations is still ongoing. Furthermore, a method able to simultaneously perform wave segmentation/delineation and denoising would be very welcome, since it would allow practitioners to rely on a single algorithm to perform these two crucial tasks required for diagnosis.

Many of the existing ECG denoising schemes are regression-based methods. These techniques consider the observed ECG signal as a noise-contaminated curve, and try to estimate the unknown clean signal from the noisy one. For example, a large class of algorithms relies on a representation of the ECG signal on a relevant, pre-defined function basis, such as wavelets. Indeed, multiresolution wavelet-based thresholding methods have been applied to denoising [10], [11], and have also been successfully used in delineation [12], [13]. In a more recent and related contribution, [14] used Variable Frequency Complex Demodulation techniques to provide an effective time-frequency decomposition of noisy ECG signals, sharing some similarities with wavelet denoising. However, regression-based methods strongly rely on a suitable choice of basis functions and carefully tuned parameters. For example, the performance of wavelet denoising methods is strongly influenced by the number of detail levels kept for reconstruction,

T. Trigano is with the Department of Electrical and Electronics Engineering, SCE, Ashdod, Israel, e-mail: thomast@sce.ac.il

S. Talala is with the Department of Electrical Engineering, Ben-Gurion University, Beer-Sheva, Israel

D. Luengo is with the Department of Audiovisual and Communications Engineering, UPM, Madrid, Spain, e-mail: luengod@ieee.org

Manuscript received September 20, 2023; revised September 20, 2023

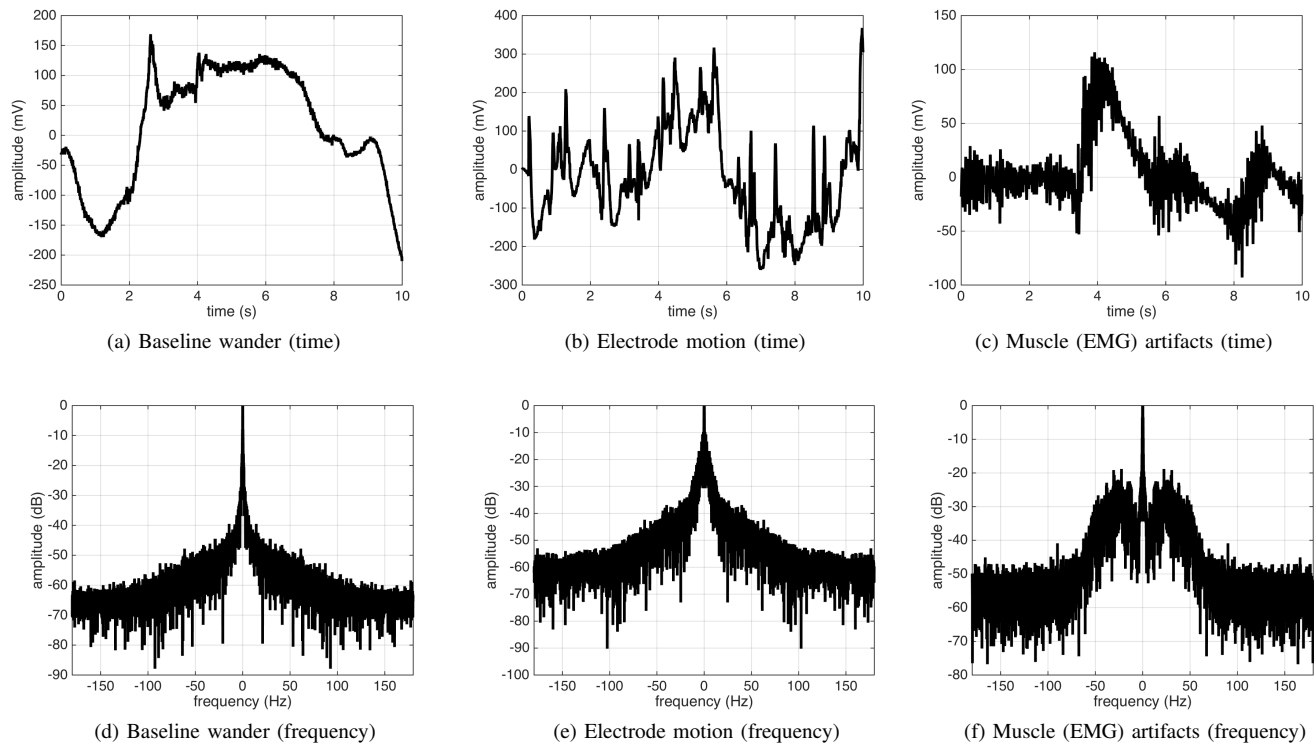


Fig. 1. Time and frequency characteristics of three of the main artifacts in ECG signals [4]: baseline wander, electrode motion, and muscle (EMG) artifacts. All the signals were taken from the first 10 seconds of the first channel of the corresponding record ('bw', 'em' and 'ma', respectively) from the MIT-BIH noise stress test database [5], as downloaded from Physionet [6].

the threshold used to remove the contribution of the noise, and the choice of the wavelet mother function. Thus, a fixed setup might yield significantly different performance from one ECG recording to another.

On the other hand, the developments in the theory of compressive sensing (CS) have also allowed researchers to provide good reconstruction and representation of the ECG signals with an excellent level of compression, suitable for real-time applications [15]–[19]. For example, [20] considers the denoising of ECG signals as a regression problem. In previous contributions, the authors have also considered ECG recordings as a sparse linear combination of Gaussian shapes to locate peaks in recorded intracardiac signals [21]–[24]. In another recent contribution, [25] suggested an iterative filtering procedure based on a group-sparse reconstruction in the frequency domain. The sparsity framework is well tailored for ECG signals, which exhibit refractory periods and are sparse by nature. However, one important shortcoming of these methods is that they rely on a known, user-defined dictionary. This requires either fine tuning or trial-and-error approaches to obtain a suitable adequacy between the data and the dictionary.

Finally, many deep learning (DL)-based approaches, based on different types of deep neural networks (DNNs), have been recently proposed for both denoising and delineation in several contexts. For example, a scheme based on a 1D convolutional neural network (CNN) for R peak detection in noisy, low-quality Holter recordings was proposed in [26], and the combination of a CNN with a Long-Short Term Memory (LSTM) network has been described in [27], [28].

Multiple DL-based methods for denoising have also been proposed, for instance, using deep convolutional encoder-decoder networks [29], CNNs [30], sophisticated deep score-based diffusion models [31], or a combination of a convolutional auto-encoder and a conditional generative adversarial network [32]. DL-based methods provide excellent results, but they require fully supervised training on large datasets to attain this performance level, and lead to heavy models from a storage/memory point of view. Hence, given the increasing amount of ECG recordings daily obtained from cardiac monitors, there is a growing need for unsupervised methods [33]. Furthermore, with the increasing number of low-cost and portable ECG monitoring devices, the need for reliable, low-power, low-cost and low-memory ECG analysis algorithms remains paramount [34].

In this paper, we present a new method (AdapTrend) for denoising and delineation based on trend filtering. This paper details three main contributions. First, we present a unified framework, based on sparse regression and trend filtering, jointly with an adaptive scheme, to both remove the noise from ECG recordings and to locate the peaks of the pulses. Besides its simplicity, the method has the additional advantage of being independent of any user-defined dictionary, in contrast to previously described ML and compressive sensing approaches. Then, we detail the implementation of the proposed method, and we provide practical rules-of-thumb to set up the hyperparameters of the algorithm. Eventually, we illustrate the good performance of the proposed approach, both on simulated and real data.

We notice that the proposed approach has three key advantages with respect to the existing techniques:

- 1) Although the proposed approach looks similar to regression-based methods, like wavelet-based denoising or CS schemes, it does not rely on any pre-defined shape. Indeed, though we add a sparsity penalty in the functional of equation (3), this sparsity is enforced in the functional regularity, and no dictionary of shapes is present in (3).
- 2) Unlike DL-based approaches, which require a large amount of labelled training data, our approach is completely unsupervised and can work with the short recordings which are acquired in some applications. Furthermore, being unsupervised it is immune to label noise and can be applied to the many unannotated ECG recordings available both in clinical and ambulatory settings.
- 3) Compared to DL-based approaches, AdapTrend requires a negligible amount of memory (compared to the millions of parameters that DNNs need to store), training time (minutes vs. hours or even days) and computational power (no GPUs or any other specific hardware required). Although developing an efficient version of AdapTrend that can be implemented in a portable ECG monitor is out of the scope of this paper, the proposed approach is clearly better suited for this setting than existing DL-based methods.

Also related in a sense to time-frequency analysis, Empirical Mode Decomposition methods have also been successfully applied, both for ECG denoising [35]–[37] and delineation [38]. This approach shares a few similarities with the proposed method, in the sense of being unsupervised and the fact that the iterative computation of the envelope of the residual is strongly related to the signal's regularity. However, from a mathematical point of view, AdapTrend is completely different, as it relies on compressive sensing ideas rather than on time-frequency analysis.

The validation of the proposed method is performed in two steps. At first, we generate synthetic signals based on the well known ECGSYN model, on which real recorded noise (baseline wander, electrode motion and EMG artifacts), taken from the MIT-BIH noise stress test database [5], available from PhysioNet [6], is added, and we compare the performances on peak location with another state-of-the-art approach (ECGDeli). Then, we investigate the performance of the denoising and delineation procedure using the real annotated signals from the LUDB [39], taken from the PhysioBank dataset [6]. In this case, the proposed approach has been tested on all the available leads (i-iii, v1-v6, avf, avl or avr) and with respect to the different heart conditions of the available subjects (sinus rhythm, AF, sinus arrhythmia and bradycardia), showing an excellent performance in all cases.

The rest of the paper is organized as follows: the next section presents the model used and the proposed algorithm. We provide an example on a synthetic signal as a proof of concept, and show how the method can be implemented in practice using the alternating direction method of multipliers (ADMM). Section III presents the results obtained on synthetic

and real data, which confirm the success of the proposed method in jointly performing signal denoising and peak localization. Finally, Section IV contains the conclusions and some possible future lines.

II. ALGORITHM OVERVIEW

In this section we provide an overview of the model we investigate, and recall the the different ℓ_1 penalized regression that will be used for noise reduction.

A. Model Description

Along the paper, we investigate the following model:

$$y[n] = s[n] + r[n] + \varepsilon[n], \quad n = 1 \dots N. \quad (1)$$

In Equation (1), $s[n]$ denotes a slowly varying signal representing the baseline of the ECG recording and the respiration noise, $r[n]$ denotes a more-rapidly varying signal representing the ECG pulses we wish to retrieve, and $\varepsilon[n]$ is an additional noise term including all the perturbations with higher frequencies (e.g., EMG and electrode motion artifacts), with the addition of a Gaussian noise with known variance. Obviously, the model described can be vectorized as

$$\mathbf{y} = \mathbf{s} + \mathbf{r} + \boldsymbol{\varepsilon} \quad (2)$$

where the vectors $\mathbf{y}$, $\mathbf{s}$, and $\boldsymbol{\varepsilon}$ are N -dimensional, denoting respectively the overall recorded signal, and the separate contributions of the baseline wander, the ECG pulses, and the higher-frequency additive noise. Using (2), the general objective of any denoising and delineation algorithm is to isolate $\mathbf{r}$ from $\mathbf{y}$, while introducing as little distortion as possible in $\mathbf{r}$, and to locate relevant features from the obtained estimate of $\mathbf{r}$. In this paper, we will focus on the denoising and peak finding (that is, locating the local maxima of the P, QRS, and T waves of the obtained recordings) tasks.

If the waveforms of the ECG signals were perfectly known, a possible approach to address the delineation problem would be to build a dictionary from these known shapes, $\mathbf{A}$, and build a sparse representation of $\mathbf{y}$ in which the non-zero entries of the regressor would indicate each waveform's location. Such methods have been investigated in previous contributions [21], [22]. However, these approaches strongly rely on the fact that the dictionary is precisely defined, which can be challenging in most biomedical applications.

Without any parametric shape assumption, we shall exploit a different specificity of the ECG signal: the fact that it is a narrow-band signal, which can be interpreted as a kind of sparsity for the signal's rate of variation. For example, the baseline varies much slower than the ECG waveforms, and the powerline interference much faster. In a given ECG pulse, the QRS complex evolves faster than the P and T waves. Thus, we might locate the waveforms from the ECG recordings using this information inside a regression model; more specifically, we wish to penalize the derivatives of the signal rather than the regressor itself. Such ideas are common in functional data analysis [40], [41], and are used in ECG delineation when approximating the recorded signals using cubic splines. However, in the references mentioned above,

the penalties introduced are usually defined by means of the ℓ_2 -norm, which does not induce sparsity. Thus, we will detail in this paper a variation of a more recent type of sparsity-inducing regressor, namely ℓ_1 trend filtering.

B. Denoising and Delineation Framework Using Adaptive Trend Filtering

Trend filtering is a relatively new method for nonparametric regression, proposed independently by [42] and [43]. Given a user-specified order $k \geq 0$, the trend filtering estimate is defined as

$$\hat{\mathbf{x}}^{\text{trend}}(\lambda) = \arg \min_{\mathbf{x}} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{x}\|_2^2 + \lambda \|\mathbf{D}^{k+1} \mathbf{x}\|_1 \right\}, \quad (3)$$

where the minimization of (3) is performed with respect to the free variable $\mathbf{x}$, and where λ denotes a user-defined parameter, which controls the trade-off between the solution's precision, appearing in the ℓ_2 -norm, and the polynomial behavior of $\hat{\mathbf{x}}^{\text{trend}}$, described by the ℓ_1 -norm. The $\mathbf{D}^{k+1} \in \mathbf{R}^{(N-k-1) \times N}$, is the discrete difference (or derivative) operator of order $k+1$, and is defined recursively as

$$\mathbf{D} = \begin{bmatrix} -1 & 1 & 0 & \dots & 0 \\ 0 & -1 & 1 & \dots & 0 \\ \vdots & & \ddots & \ddots & \\ 0 & 0 & \dots & -1 & 1 \end{bmatrix}$$

and $\mathbf{D}^{k+1} = \mathbf{D}\mathbf{D}^k$ for $k \geq 1$. As shown in [43], the penalty term $\|\mathbf{D}^k \mathbf{x}\|_1$ introduced in (3) enforces the solution $\hat{\mathbf{x}}^{\text{trend}}(\lambda)$ to exhibit a piecewise polynomial behavior, where the number of knots between the different pieces decreases as λ increases, and where the order of the polynomials appearing in the reconstruction is defined by k (for example, setting $k = 0$ provides a piecewise linear solution, and so forth). Contrary to standard sparse regression techniques, we can see trend filtering as a sparse penalty applied on the functional smoothness of the result. This property makes it suitable to smooth noisy data, and to estimate the localization of portions of signals with different rates of variation. In this paper, since the relevant waveforms in ECG recordings are usually very smooth, we found $k = 1$ to be a good compromise to properly model the desired P-QRS-T waves and avoid overfitting (i.e., capturing undesired oscillations due to noise and artifacts).

The approach we suggest is summarized in Figure 2. At first, the regular trend filtering algorithm is used to remove the baseline wander and any low-frequency perturbations from the noisy signal $\mathbf{y}$. Then, we perform the iterations of the trend filtering optimization, but with time-varying parameters (see the later part detailing the implementation of the optimization solver). In the following, we denote this method as AdapTrend, for *adaptive trend filtering*. In practice, both stages can be solved numerically using the well-known ADMM algorithm; as detailed later on, this algorithm depends on the choice of a (possibly varying) parameter $\rho[n]$. During this stage, we are interested in performing several iterations of adaptive trend filtering, which will allow us to estimate the fiducial points' locations based on these iterations. At each iteration, the position of the local maxima of the current estimate

of $\mathbf{r}$ are recorded. Based on these local maxima and the residual obtained after all the ADMM iterations, we first derive the position of the QRS complexes. Then, based on this information and the remaining local maxima found, we derive the position of the P and T waves extrema. Though the theoretical study of this estimator is beyond the scope of this contribution, we can advance some intuitive explanation in defense of AdapTrend. First, we observe that QRS complexes are narrow peaks, which have little influence on the term $\|\mathbf{y} - \mathbf{x}\|_2^2$ appearing in (3). Thus, this part of the ECG signal can be easily detected in the residual $\mathbf{y} - \hat{\mathbf{s}} - \hat{\mathbf{x}}^{\text{trend}}(\lambda)$ in the last step of the algorithm. Furthermore, making the trend filtering parameter vary has a direct influence on the number of knots appearing in the solution. More specifically, as the parameter ρ , defined later on in the implementation, decreases, the number of intervals in which the solution is piecewise polynomial increases. Thus, by choosing large values for both λ and ρ , as detailed later, we can obtain a baseline which does not take into account the ECG pulses. However, making ρ vary and decrease (as already mentioned, for example, in [43]) with the ADMM iterations allows us to detect the waves of the ECG pulses recursively, based on the QRS location. Indeed, since the obtained solution is piecewise polynomial, the local extrema of the pulse's waves can be extracted from the zeroes of the numerical differentiation of the obtained solution for each $\rho[n]$.

To illustrate this intuitive explanation, we present the results obtained step by step on an artificial signal generated by means of the ECGSYN model [44], [45]. Figure 3 shows the original signal and the baseline obtained after the first trend filtering step. Note that up to a constant (basically, the signal's average), the baseline obtained is close to a sine wave, which is consistent with the simulated baseline (a slowly varying sine) in the ECGSYN model.

At each iteration of the algorithm, we record the local maxima obtained on $\hat{\mathbf{x}}^{\text{trend}}(\lambda)$. The results are displayed in Figure 4. Indeed, since trend filtering tends to minimize the ℓ_2 -norm, it will tend at first to retrieve the peaks providing the largest contribution in this sense. In the case of the specific simulated signal at hand, since the T-waves have both a large support and a relatively high amplitude, the reconstruction will start with them. QRS complexes will follow, and eventually the contribution of the P-waves will be acknowledged, as shown in Fig. 5. Finally, the residual at the end of the process (100 iterations in our case) is an excellent signal to locate the QRS complexes, as shown in Fig. 6.

We now summarize AdapTrend in a more formal manner. We denote by $m_i[n]$ the sample path of the local maxima locations at iteration i (that is $m_i[n] = 1$ if a local maxima is present at the sampling time n , and 0 otherwise), and $\ell_{\min} < \ell_{\max}$ are two integers. We also define the history of the local maxima as

$$m[n] = \sum_{i=\ell_{\min}}^{\ell_{\max}} m_i[n], \quad (4)$$

In the previous sum, the parameters $\ell_{\max}$ is introduced to acknowledge that trend filtering might estimate the additive

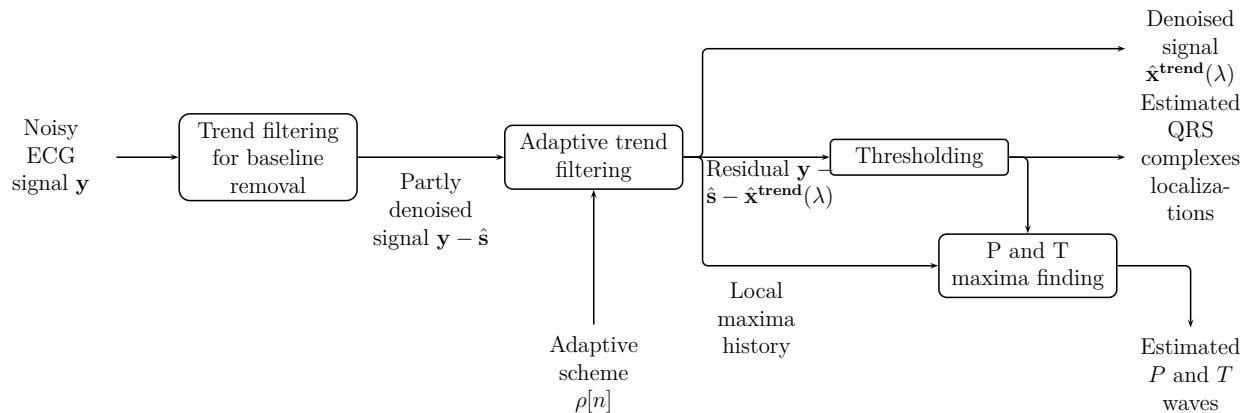


Fig. 2. Proposed framework for ECG denoising and fiducial points extraction.

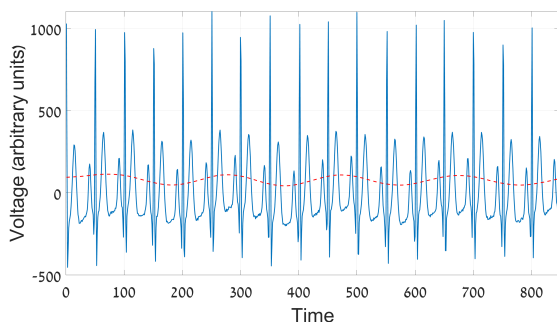


Fig. 3. Baseline obtained using the trend filtering, with $\lambda = 1000$ and $\rho = 100$.

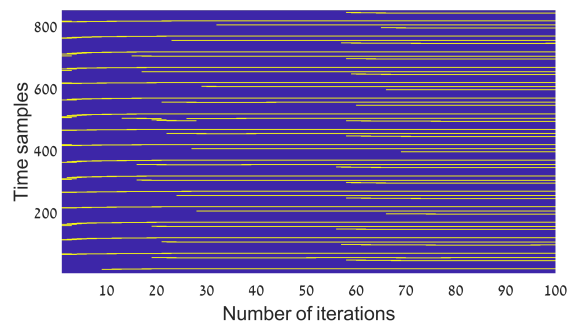


Fig. 4. Local maxima positions obtained at each iteration of the trend filtering with the given $\rho[n]$. From iteration 70 and beyond, the peaks of the P, QRS and T waves are well located.

noise for small values of $\rho[n]$, and ℓ_{min} accounts for the “burn-in” of the preliminary iterations, as explained later on.

The delineation procedure starts with the localization of the QRS complexes. They are selected based on a threshold applied on the residual $y - \hat{s} - \hat{x}^{\text{trend}}(\lambda)$. More specifically, all the local extrema above thrice the average of the values of $y - \hat{s} - \hat{x}^{\text{trend}}(\lambda)$ are selected as QRS complexes. We denote the sampling points selected as the maxima of the QRS

complexes by $\{q_1, \dots, q_M\}$. We then define the k -th anchor point in the local maxima’s history $m[n]$ as

$$a_k = \arg \min_{j \in \mathcal{C}_m} |j - q_k|, \quad 1 \leq k \leq M$$

where $\mathcal{C}_m$ is the set of the local maxima present in $m[n]$. Then, the T and P waves extrema are estimated based on the history of the trajectories and the QRS locations. The k -th T-wave and k -th P-wave extrema, denoted by t_k and p_k , respectively, are estimated as

$$t_k = q_k + \text{off}_T + \arg \min_{j \in \mathcal{C}_m, a_k + \text{off}_T < j < a_{k+1}} |j - a_k|$$

$$p_k = q_k - \text{off}_P + \arg \min_{j \in \mathcal{C}_m, a_{k-1} < j < a_k - \text{off}_P} |j - a_k|.$$

The additional offsets off_T and off_P are introduced in the latter to take into account physiological constraints and ensure that the retained local extrema are not too close to one from the other. In practice, taking a number of sampling points equivalent to $0.2s$ for off_P and off_T provided satisfactory results, and we used this value for all the results presented in this paper.

The use of ℓ_{min} and ℓ_{max} in (4) is of little importance for the example presented before, since the synthetic signal is noise-free. However, these parameters are critical to obtain satisfactory results on real, noisy recordings. To illustrate this point, we present in Figure 7 the obtained local maxima for a small part of a noisy recording taken from the LUDB dataset (further details and results on this dataset appear in the last section of the paper). At the early stages of AdapTrend, for large values of $\rho[n]$, the algorithm does not take into account spiky parts of the signals, since they do not contribute significantly to the minimization of the ℓ_2 -norm in (3). Thus, the obtained local maxima change from one iteration to another and do not represent ECG fiducial points. On the other hand, at the latest stages, the algorithm tends to overfit; thus, we observe numerous local maxima detected due to the noise. In light of this example, we understand that it is necessary for real data to

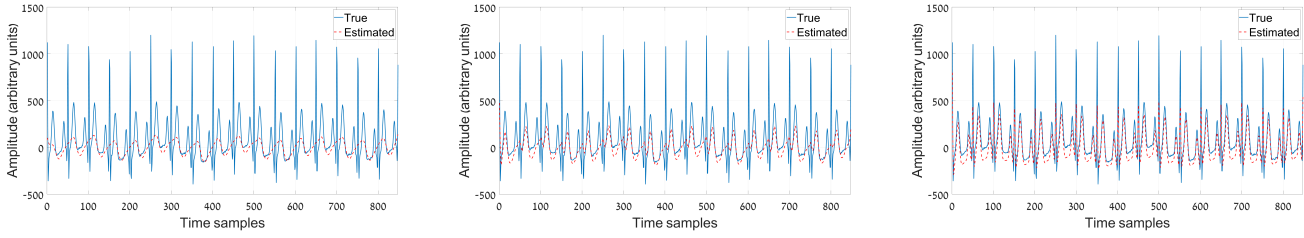


Fig. 5. Examples of reconstructed signals (including baseline removal) at iterations 10 (up), 50 (center) and 100 (down) using a varying sequence of $\rho[n]$ according to (11).

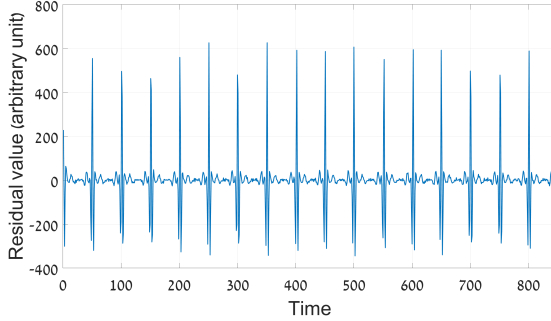


Fig. 6. Residual obtained after 100 iterations of AdapTrend with $\lambda = 1000$ and a varying sequence of $\rho[n]$ according to (11).

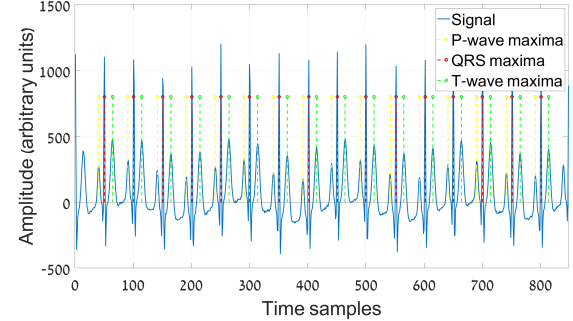


Fig. 8. Local maxima positions obtained at each iteration of the trend filtering with a decreasing $\rho[n]$ as described in (11), $\ell_{\min} = 1$ and $\ell_{\max} = 100$. Note that all the P, QRS and T waves are well located.

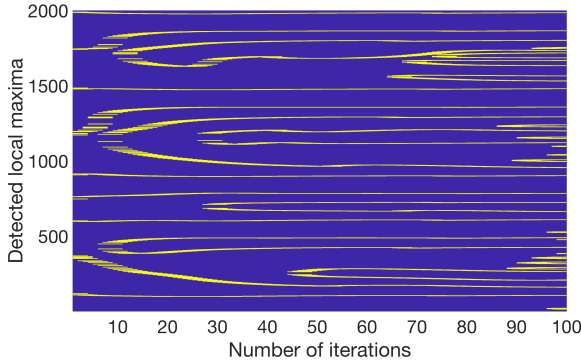


Fig. 7. Example of local maxima obtained on a noisy recording.

discard the local maxima obtained in early and late iterations of AdapTrend in (4), to locate ECG peaks more precisely. A good practical rule-of-thumb that gave satisfactory results in our experiments is to keep about 60% of the iterations, in-between the initial and the final step, for the computation of $m[n]$. However, as seen in the later section, this percentage can be higher for noise-free signals or recordings with very high SNR.

Results on this specific signal are displayed in Figure 8. On this specific example, it seems that trend filtering is enough to address both the denoising and delineation problem for ECG recordings, and this procedure is both simple and fast. Before trying to validate the approach at a larger level, we now discuss its practical implementation using the alternating direction method of multipliers (ADMM).

C. Practical Implementation

ADMM is a simple but powerful algorithm that is well suited to distributed convex optimization in general, and in particular to problems arising in applied statistics and machine learning. ADMM solves problems by breaking them into smaller pieces, each of which is then easier to handle. The algorithm solves problems in the form

$$\begin{aligned} & \text{minimize} && f(\mathbf{x}) + g(\mathbf{z}) \\ & \text{subject to} && \mathbf{Ax} + \mathbf{Bz} = \mathbf{c} \end{aligned} \quad (5)$$

where $\mathbf{x} \in \mathbf{R}^n, \mathbf{z} \in \mathbf{R}^m, \mathbf{c} \in \mathbf{R}^p$ are vectors and $\mathbf{A} \in \mathbf{R}^{p \times n}, \mathbf{B} \in \mathbf{R}^{p \times m}$ are matrices. To solve this problem we need to replace the problem with a similar problem as follows:

$$\begin{aligned} & \text{minimize} && f(\mathbf{x}) + g(\mathbf{z}) + \frac{\rho}{2} \|\mathbf{Ax} + \mathbf{Bz} - \mathbf{c}\|_2^2 \\ & \text{subject to} && \mathbf{Ax} + \mathbf{Bz} - \mathbf{c} = \mathbf{0} \end{aligned} \quad (6)$$

This problem is clearly equivalent to the original problem (5), since for any feasible $\mathbf{x}, \mathbf{z}$ the term added to the objective is zero. The augmented Lagrangian for problem (6) is

$$\begin{aligned} L_\rho(\mathbf{x}, \mathbf{z}, \mathbf{u}) = & f(\mathbf{x}) + g(\mathbf{z}) + \mathbf{u}^T (\mathbf{Ax} + \mathbf{Bz} - \mathbf{c}) \\ & + \frac{\rho}{2} \|\mathbf{Ax} + \mathbf{Bz} - \mathbf{c}\|_2^2 \end{aligned} \quad (7)$$

where $L_\rho(\mathbf{x}, \mathbf{z}, \mathbf{u})$ is the augmented Lagrangian, and $\rho > 0$ is a user-defined parameter, possibly varying with the iterations. ADMM finds the candidate points in (7) iteratively by sequentially solving the following equations:

$$\begin{cases} \mathbf{x}^{(n+1)} &= \arg \min_{\mathbf{x}} L_\rho(\mathbf{x}, \mathbf{z}^{(n)}, \mathbf{u}^{(n)}) \\ \mathbf{z}^{(n+1)} &= \arg \min_{\mathbf{z}} L_\rho(\mathbf{x}^{(n+1)}, \mathbf{z}, \mathbf{u}^{(n)}) \\ \mathbf{u}^{(n+1)} &= \mathbf{u}^{(n)} + \rho(\mathbf{Ax}^{(n+1)} + \mathbf{Bz}^{(n+1)} - \mathbf{c}) \end{cases} \quad (8)$$

where $\mathbf{x}^{(n)}$, $\mathbf{z}^{(n)}$ and $\mathbf{u}^{(n)}$ denote the values of the minimizers at iteration n . ADMM can be interpreted as iterating the optimization one variable after the other in (8), until the algorithm converges to a local minimizer.

In the case of trend filtering, the problem in (5) can be rewritten as

$$\begin{aligned} & \text{minimize} && \|\mathbf{y} - \mathbf{x}\|_2^2 + \lambda \|\mathbf{z}\|_1 \\ & \text{subject to} && \mathbf{D}^{k+1} \mathbf{x} - \mathbf{z} = \mathbf{0}, \end{aligned}$$

so the iterations become (see [46] for additional details):

$$\begin{cases} \mathbf{x}^{(n+1)} &= (\mathbf{I} + \rho(\mathbf{D}^{k+1})^T \mathbf{D}^{k+1})^{-1} \\ &\quad \times (\mathbf{y} + \rho(\mathbf{D}^{k+1})^T (\mathbf{z}^{(n)} - \mathbf{u}^{(n)})) \\ \mathbf{z}^{(n+1)} &= \mathbf{S}_{\lambda/\rho}(\mathbf{D}^{k+1} \mathbf{x}^{(n+1)} + \mathbf{u}^{(n)}) \\ \mathbf{u}^{(n+1)} &= \mathbf{u}^{(n)} + \mathbf{D}^{k+1} \mathbf{x}^{(n+1)} - \mathbf{z}^{(n+1)} \end{cases} \quad (9)$$

where $\mathbf{S}_x$ in (9) is the soft thresholding operator, defined by

$$\mathbf{S}_x(a) = \begin{cases} a - x, & a > x; \\ 0, & |a| \leq x; \\ a + x, & a < -x. \end{cases}$$

In practice, this optimization procedure depends on two free parameters, λ and ρ . [47] suggested a way to estimate the largest λ parameter that will give us good convergence, as well as the Lagrangian parameter ρ , for regular trend filtering. We will use this choice for the first step of the proposed method, since its objective is the baseline removal. More specifically, we set $\lambda = \rho = \lambda_{max}$, where the parameter λ_{max} is defined by

$$\lambda_{max} = \|((\mathbf{D}^{(k+1)}(\mathbf{D}^{(k+1)})^T)^{-1}(\mathbf{D}^{(k+1)})^T \mathbf{y})\|_\infty. \quad (10)$$

For example, the baseline removal in Figure 3 was obtained using this choice of parameters. The choice of the sequence parameter $\rho[n]$ should also be made carefully: small values of $\rho[n]$ provide a trend filtering regressor $\mathbf{x}$ close to the original signal $\mathbf{y}$, at the expense of its smoothness. On the contrary, when $\rho[n]$ is large, we obtain a very smooth solution, which does not reflect the presence of sharp peaks such as QRS complexes. Since the peaks we aim to retrieve have different dynamics, we suggest in this paper to define $\rho[n]$ as a slowly-decreasing sequence. The results obtained in Figures 5 and 8, for example, were obtained with 100 iterations of the ADMM and

$$\rho[n] = 10^{x_n}, \quad x_n = 2 - \frac{3n}{100}, \quad n = 0 \dots 100. \quad (11)$$

We shall later on use sequences of this type in our simulations and applications on real-life datasets.

III. RESULTS

The results obtained for delination of ECG recordings, both on synthetic and real signals, are presented in this section.

A. Simulations Overview

As shown in the algorithm description, the ECGSYN introduced in [44] provides regular, almost noise-free, simulated ECG signals. We illustrated in the previous section the performances of the proposed approach on this type of

signals, showing the excellent location ability of our proposed AdapTrend method on clean signals in Figure 8. However, a simulation uniquely based on them would provide little insight about the algorithm's shortcomings.

Therefore, the simulations performed are described as follows. We draw 1000 simulated ECGs of $N = 1000$ points and $N = 5000$ points, based on the ECGSYN model. For each generated signal, noise is randomly added in some parts of the signal, this noise being either from a real baseline wander, from electrode motion or from EMG. The noise recordings are taken from the MIT-BIH noise stress test database [5], [6]. For each simulated example, we investigate the algorithm's robustness to the different kinds of noise for the following SNRs: 20 dB, 12 dB, 10 dB, 6 dB, 3 dB, 0 dB, -3 dB and -6 dB. This robustness is evaluated by means of the positive predictive value (PPV), which is evaluated separately for each type of waves (P, QRS and T) as

$$\begin{aligned} PPV(M) &= \frac{\#\{k \ ; \exists m \ ; \ |x_k - \hat{x}_m| < M\}}{\#\{k \ \exists m \ ; \ |x_k - \hat{x}_m| < M\} + \#\{m \ \forall k \ ; \ |x_k - \hat{x}_m| \geq M\}}, \end{aligned} \quad (12)$$

where $\#A$ denotes the cardinality of the set A , x_k is the true position of the k -th peak, $\hat{x}_m$ is the estimated position of the m -th peak and $M - 1$ is the maximum absolute deviation allowed between the true and the estimated peak positions. Based on (12), we decide that a peak was well located if the algorithm provides a location close to a true one, up to 10 samples. The method presented in this paper is then compared to the results obtained for the same signals with the ECGDeli Matlab library [48], which provides excellent results on real signals for delineation of multichannel ECG recordings. The parameters in AdapTrend were set up as follows: for the iterations, the parameter λ was set based on (10). For the baseline removal part of AdapTrend, we used a constant $\rho = 100$ in the ADMM, and ran 100 iterations. For the peak localization step, we ran also 100 iterations of the ADMM, while we set the parameter $\rho[n] = 10^{-1+3n/100}$. Since the signals obtained with the ECGSYN model are noise-free (besides our manual addition), we can rely on most of the history of the local maxima, and chose for (4) $l_{min} = 20$ and $l_{max} = 100$.

B. Results on Simulations

As a visual example, we present results obtained for baseline wander, EMG noise and electrode motion in Figures 9, 10 and 11, respectively. It can be noted that the localization of all the relevant peaks is excellent, though we observe a predictable decrease in performance when dealing with electrode motion noise.

The global results for QRS complexes, P-wave and T-wave detections obtained with different SNRs for signals of size $N = 1000$ points are displayed, for both algorithms (AdapTrend and ECGDeli), in Figures 12, 13 and 14, respectively. The results obtained on signals of size $N = 5000$ are presented in figures 15, 16 and 17.

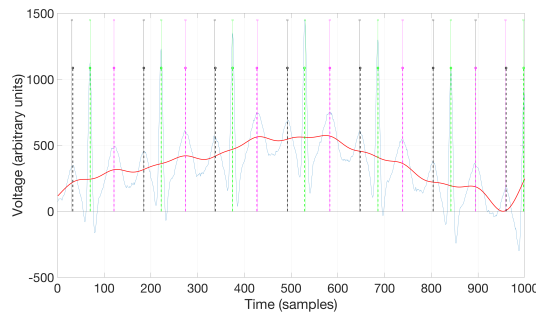


Fig. 9. Example of peak localization for a simulated ECG with baseline wander (blue) - true positions are in dashed bold colors, and estimated locations are in plain colors. Estimated baseline is displayed in red.

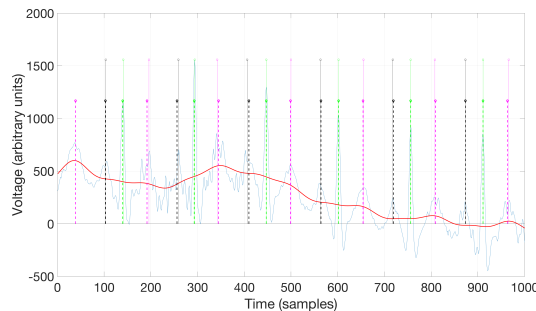


Fig. 10. Example of peak localization for a simulated ECG with EMG noise (blue) - true positions are in dashed bold colors, and estimated locations are in plain colors. Estimated baseline is displayed in red.

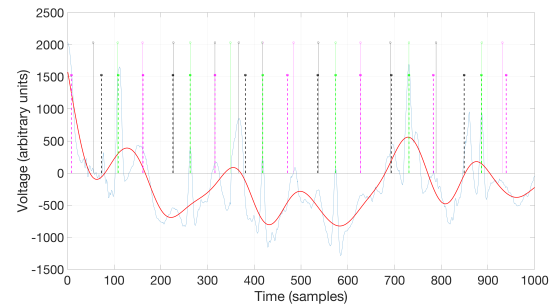


Fig. 11. Example of peak localization for a simulated ECG with electrode motion noise (blue) - true positions are in dashed bold colors, and estimated locations are in plain colors. Estimated baseline is displayed in red.

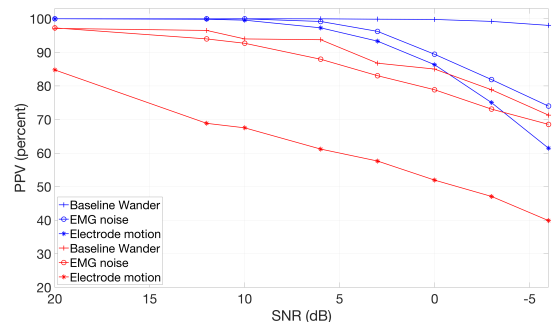


Fig. 12. Performances in terms of $PPV(10)$ for QRS complexes localization on ECGSYN signals ($N = 1000$) with different kinds of additive noises. Blue: AdapTrend. Red: ECGDeli

C. Results on Real Data

We present the obtained results on the Lobachevsky University Electrocardiography Database (LUDB), available on Physionet [39]. LUDB includes recordings with marked boundaries and peaks of P, T waves and QRS complexes. The database consists of 200, 10-second, 12-lead ECG signal recordings representing different morphologies of the ECG signal, sampled at 500 Hz and manually annotated. The ECGs were collected from healthy volunteers and patients suffering from various cardiovascular diseases at the Novgorod Hospital in 2017-2018. We evaluated the performances of the proposed approach on all the recordings, and for all channels. However, it should be noticed that one particularity of the LUDB dataset is that the first and last seconds of the recordings are not manually annotated, thus the quantitative evaluation of our algorithm cannot be performed on these parts of the signals. Furthermore, the signal is not annotated where no peak is visually distinguishable from the baseline. For this, we performed for both parts of AdapTrend 100 iterations of the ADMM. For the peak localization, we set $\rho[n] = 10^{2+6n/100}$, $n = 0 \dots 99$, $l_{min} = 20$ and $l_{max} = 80$.

We present in Figures 18, 19, 20, 21 and 22, some results obtained on different recordings, for patients with variable heart conditions (atrial fibrillation, sinus arrhythmia, sinus rhythm, sinus bradycardia and sinus tachycardia, respectively). Lead iii was chosen due to its resemblance with the artificial signals generated by the ECGSYN model in the simulation sections; as shown in Table I, however, this lead is not

necessarily related with the most precise results. These figures illustrate that the proposed method is relatively robust to different heart conditions, even when some peaks (particularly the P-waves) are hard to detect in practice. Furthermore, even with the lack of annotations at the signals' tails, it can be observed visually that the proposed approach performs quite well on the unannotated parts of the signals.

Lead	P-wave peak localization	QRS localization	T-wave peak localization
i	0.9709	0.99293	0.94303
ii	0.97445	0.99728	0.95152
iii	0.94543	0.98152	0.94286
v1	0.98862	0.98424	0.94165
v2	0.95516	0.97444	0.93489
v3	0.96165	0.98152	0.95455
v4	0.96793	0.98912	0.94909
v5	0.95878	0.98586	0.93277
v6	0.94026	0.97009	0.92611
avf	0.92832	0.98370	0.94852
avl	0.92629	0.99402	0.95581
avr	0.95529	0.98586	0.96002

TABLE I

PEAK LOCALIZATION RESULTS ($PPV(60)$) OBTAINED ON THE LUDB DATASET, FOR EACH LEAD PROCESSED INDIVIDUALLY

In order to quantitatively assess the performances of the proposed approach, we follow the methodology described in [9] and assume that a difference greater than 120 ms (60 sampling points for the LUDB dataset) between the annotation

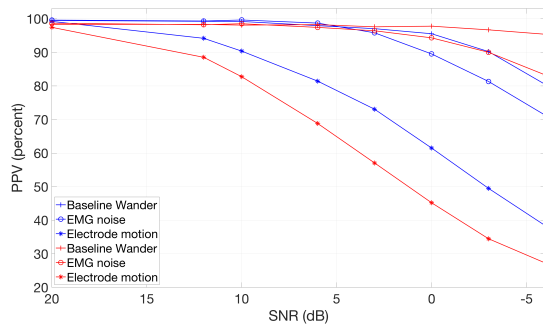


Fig. 13. Performances in terms of $PPV(10)$ for P-wave localization on ECGSYN signals ($N = 1000$) with different kinds of additive noises. Blue: AdapTrend. Red: ECGDeli

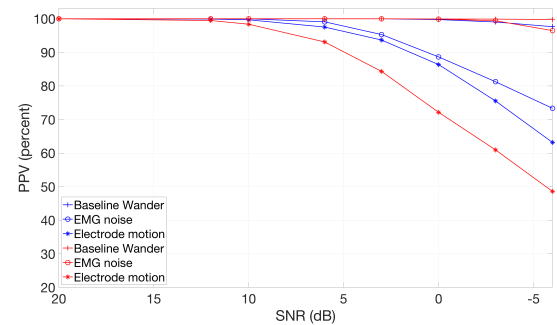


Fig. 15. Performances in terms of $PPV(10)$ for QRS complexes localization on ECGSYN signals ($N = 5000$) with different kinds of additive noises. Blue: AdapTrend. Red: ECGDeli

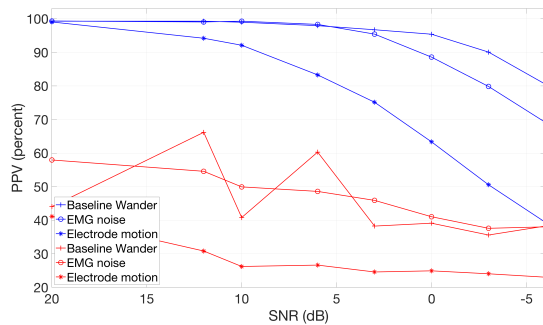


Fig. 14. Performances in terms of $PPV(10)$ for T-wave localization on ECGSYN signals ($N = 1000$) with different kinds of additive noises. Blue: AdapTrend. Red: ECGDeli

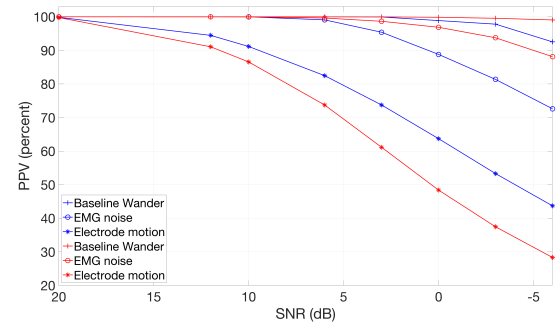


Fig. 16. Performances in terms of $PPV(10)$ for P-wave localization on ECGSYN signals ($N = 5000$) with different kinds of additive noises. Blue: AdapTrend. Red: ECGDeli

and the output provided by our method is considered to be a missed detection. This allows us to compare the obtained results with other existing methods described in [9]. The obtained results are summarized in Table I. We summarize, for the sake of comparison, the results obtained by other state-of-the-art methods, described in [9], in Table II. Note that the comparison is only partial, since all the concurrent methods were tested on the QT database, which contains longer and cleaner signals than LUDB; however, the performances of the method proposed in the paper fall within the range of the others. Thus, AdapTrend appears to be a valid alternative for peak localization. Furthermore, the most competitive methods appearing in the table strongly rely on wavelet decompositions, and thus we can expect a degradation in their performances with shorter signals, such as those in the LUDB database.

D. Discussion

From the results obtained on the ECGSYN model, it appears that AdapTrend outperforms ECGDeli in most of the cases for short signals ($N = 1000$ points). Indeed, regarding the QRS detection task, AdapTrend is better than ECGDeli for all SNRs. It is also interesting to compare these results with the well-known Pan-Tompkins method. Though the comparison does not rely on the same signals, it is remarkable that the obtained results for QRS detection are always very close to the ones obtained in [55, Table 3] for Pan-Tompkins. This seems to indicate that the proposed algorithm is a valid alternative

to Pan-Tompkins for QRS detection, with the theoretical advantage of relying on a single framework for full wave delineation. Though further investigations on similar signals should be done, this comparison alone is very encouraging.

For longer signals ($N = 5000$ points), performances are more balanced, with ECGDeli performing better than the proposed method against EMG noise, while this tendency is reversed for electrode motion noise; for baseline wander, both approaches perform more or less similarly. The most interesting point to notice is that AdapTrend is very robust to changes in N , while ECGDeli needs lots of sampling points to delineate the recorded signal efficiently; indeed, ECGDeli returns significantly better results for $N = 5000$ than for $N = 1000$. In contrast, AdapTrend results are invariant with respect to N . This last point can be explained by the methodology used in ECGDeli; since it relies on a discrete wavelet decomposition to locate pulses, longer signals yield more decomposition levels, and therefore a more precise result. On the other hand, AdapTrend relies on the functional regularity of the signal to reconstruct, rather than on the signal's length. This is a crucial advantage for the short ECG recordings used in many practical applications (e.g., single beat ECG-based biometric recognition [56] or beat-to-beat sensing in implantable devices [57]).

In the case of P-wave detection, we notice that for very low SNRs, ECGDeli outperforms the presented approach by about 10% of precision; however, both algorithms behave similarly for a wide range of signal to noise ratios for baseline wander

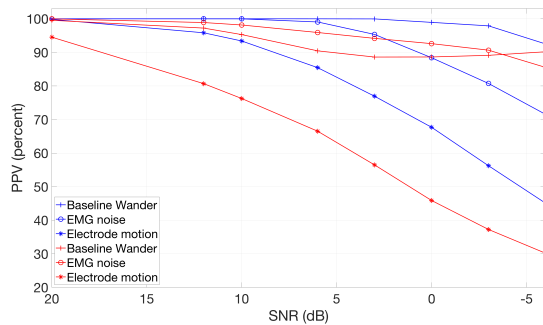


Fig. 17. Performances in terms of $PPV(10)$ for T-wave localization on ECGSYN signals ($N = 5000$) with different kinds of additive noises. Blue: AdapTrend. Red: ECGDeli

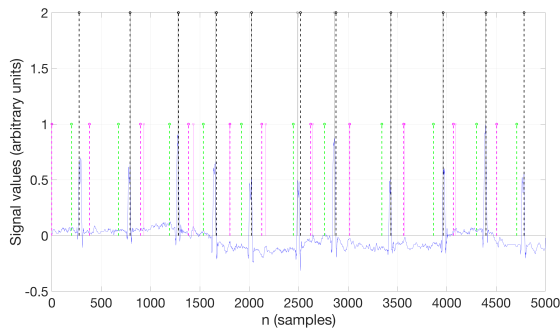


Fig. 18. Peak localization on lead *iii* for an AF patient. The estimated positions appear in dotted bold lines, and the manual annotations in plain lines.

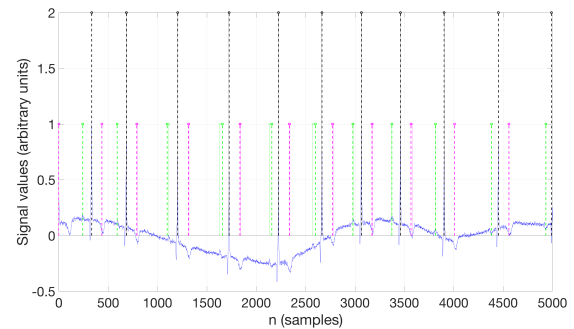


Fig. 19. Peak localization on lead *iii* for a sinus arrhythmia patient. The estimated positions appear in dotted bold lines, and the manual annotations in plain lines.

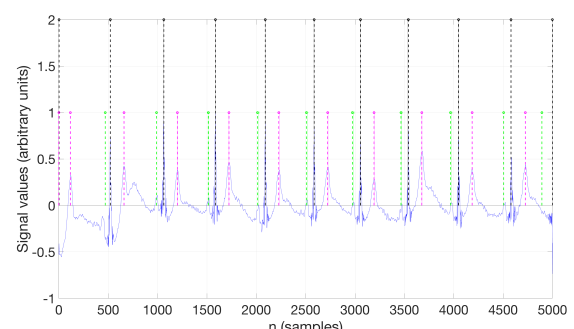


Fig. 20. Peak localization on lead *iii* for a sinus rhythm patient. The estimated positions appear in dotted bold lines, and the manual annotations in plain lines.

and EMG noise, and our algorithm performs substantially better in the case of electrode motion. For QRS detection, AdapTrend's performance is substantially better than ECGDeli for $N = 1000$, as already mentioned above. For $N = 5000$, AdapTrend still performs better for EMG noise and similar results are obtained for baseline wander, but ECGDeli outperforms AdapTrend for EMG noise. Finally, for T-wave detection AdapTrend performs much better than ECGDeli in general. Indeed, the T-wave detection yielded surprisingly poor results for ECGDeli with $N = 1000$. This issue is investigated further in the Appendix.

In summary, based on the results obtained on the simulated data, we note that the proposed method provides very good localization capabilities for both P and T waves, with PPV higher than 90% for SNRs as low as 0 dB for baseline wander and EMG noise. Not surprisingly, we note a degradation in the precision with electrode motion, while remaining reasonable. A similar conclusion can be reached with respect to QRS detection: AdapTrend attains very good results for baseline wander and EMG noise, both for shorter and longer signals. Once more, a performance degradation is appreciated for electrode motion, though not as much as that of ECGDeli.

Regarding the results obtained on clinical recordings, using the LUDB dataset, the overall the results (shown in Table I) are quite good. Indeed, one important point to emphasize is that AdapTrend is quite robust, both with respect to the lead used (i-iii, v1-v6, avf, avl or avr) and with respect to the different heart conditions tested (sinus rhythm, AF, sinus

arrhythmia and bradychardia). When comparing the leads i, ii and iii to the results obtained in [9], we can observe that the peak detection results obtained with AdapTrend fall within the range of the performances displayed by all the other state-of-the-art algorithms, which makes it relevant with respect to alternative methods. Furthermore, AdapTrend allows us to remove the noise from the recording quite efficiently, as seen for example in Figure 23, and the results on peak localization remain quite satisfactory even for very noisy signals, as seen from Figure 24.

IV. CONCLUSION

In this paper, we have presented AdapTrend, an algorithm based on ℓ_1 Trend Filtering to automatically remove additive noise from ECG recordings and jointly perform peak localization. We have showed that though simple, the proposed method performed very well on simulations, and attained localization rates identical to most existing methods in the field on real data. Furthermore, the results obtained on the LUDB dataset show that the method is quite robust to recording conditions and heart rate variability. Future contributions will focus on the peak limits localization and the theoretical study of AdapTrend, to propose optimal decreasing rates for its adaptive parameters. The robustness of AdapTrend to different heart conditions and the resulting retention of pathological ECG waveforms will also be part of this future work.

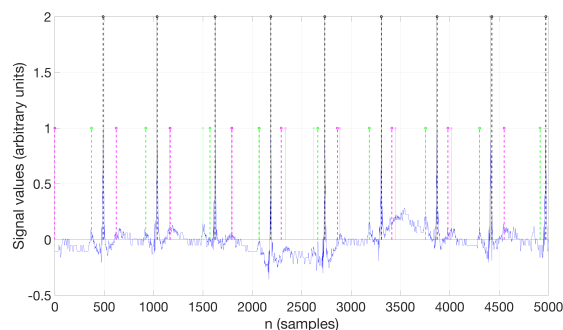


Fig. 21. Peak localization on lead *III* for a sinus bradycardia patient. The estimated positions appear in dotted bold lines, and the manual annotations in plain lines.

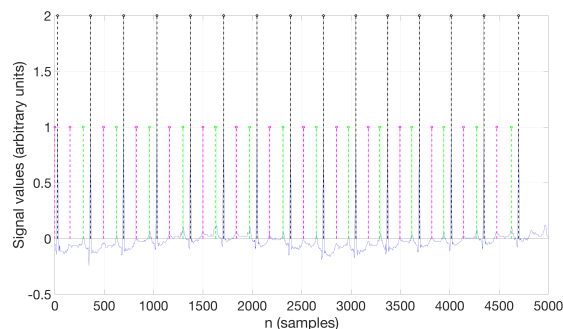


Fig. 22. Peak localization on lead *III* for a sinus tachycardia patient. The estimated positions appear in dotted bold lines, and the manual annotations in plain lines.

REFERENCES

- [1] L. Sornmo and P. Laguna, *Bioelectrical Signal Processing in Cardiac and Neurological Applications*. Elsevier, 2005.
- [2] I. Christov, "A review of electrocardiogram filtering," *Journal of Electrocardiology*, vol. 43, no. 1, pp. 486–496, 2010.
- [3] P. Laguna, R. Jane, and P. Caminal, "Automatic detection of wave boundaries in multilead ECG signals: validation with the CSE database," *Computers and Biomed. Research*, vol. 27, no. 1, pp. 45–60, Feb. 1994.
- [4] D. Luengo, D. Osés, and T. Trigano, "Digital signal processing of ECG and PCG signals," in *Teaching and Subjects on Bio-Medical Eng.*, 2021.
- [5] G. B. Moody, W. E. Muldrow, and R. G. Mark, "A Noise Stress Test for Arrhythmia Detectors," *Computers in Cardiology*, vol. 11, pp. 381–384, 1984.
- [6] A. Goldberger, L. Amaral, L. Glass, J. Hausdorff, P. Ivanov, R. Mark, J. Mietus, G. Moody, C. Peng, and H. Stanley, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals," *Circulation*, vol. 101, no. 23, pp. e215 – e220, 2000.
- [7] K. T. Sweeney, T. E. Ward, and S. F. McLoone, "Artifact removal in physiological signals—practices and possibilities," *IEEE Trans. on Information Tech. in Biomedicine*, vol. 16, no. 3, pp. 488–500, 2012.
- [8] S. Chatterjee, R. S. Thakur, R. N. Yadav, L. Gupta, and D. K. Raghuvanshi, "Review of noise removal techniques in ECG signals," *IET Signal Processing*, vol. 14, no. 9, pp. 569–590, 2020.
- [9] I. Beraza and I. Romero, "Comparative Study of Algorithms for ECG Segmentation," *Biomedical Signal Processing and Control*, vol. 34, pp. 166–173, 2017.
- [10] O. Sayadi and M. B. Shamsollahi, "Multiadaptive bionic wavelet transform: Application to ECG denoising and baseline wandering reduction," *EURASIP Journal of Adv. in Signal Proc.*, vol. 2007, pp. 1–11, 2007.
- [11] L. Smital, M. Vitek, J. Kozumplik, and I. Provaznik, "Adaptive wavelet Wiener filtering of ECG signals," *IEEE Transactions on Biomedical Engineering*, vol. 60, no. 2, pp. 437–445, Feb. 2013.
- [12] J. Martinez, R. Almeida, S. Olmos, A. Rocha, and P. Laguna, "A wavelet-based ECG delineator: evaluation on standard databases," *IEEE Trans. on Biomedical Eng.*, vol. 51, no. 4, pp. 570–581, Apr. 2004.
- [13] M. Vitek, J. Hrubec, and J. Kozumplik, "A wavelet-based ECG delineation in multilead ECG signals: Evaluation on the CSE database," in *World Congress on Medical Physics and Biomedical Engineering*, 2009, pp. 177–180.
- [14] B. Hossain, S. Khairul Bashar, J. Lazaro, N. Reljin, Y. Noh, and K. H. Chon, "A robust ECG denoising technique using variable frequency complex demodulation," *Computer Methods and Programs in Biomedicine*, vol. 200, p. 105856, 2021.
- [15] Z. Zhang, T.-P. Jung, S. Makeig, and B. Rao, "Compressed sensing for energy-efficient wireless telemonitoring of noninvasive fetal ECG via block sparse Bayesian learning," *IEEE Transactions on Biomedical Engineering*, vol. 60, no. 2, pp. 300–309, Feb. 2013.
- [16] I. Ciocoiu, "Single channel fetal ECG recovery using sparse redundant representations," in *10th Int. Symp. on Signals, Circuits and Systems (ISSCS)*, June 2011, pp. 1–4.
- [17] A. Adler, M. Elad, Y. Hel-Or, and E. Rivlin, "Sparse coding with anomaly detection," in *IEEE Int. Workshop on Machine Learning for Signal Processing (MLSP)*, Sept. 2013, pp. 1–6.
- [18] L. Polania and K. Barner, "Multi-scale dictionary learning for compressive sensing ECG," in *IEEE Digital Signal Processing and Signal Processing Education Meeting (DSP/SPE)*, Aug. 2013, pp. 36–41.
- [19] A. Ghaffari, H. Palangi, M. Babaie-Zadeh, and C. Jutten, "ECG denoising and compression by sparse 2d separable transform with overcomplete mixed dictionaries," in *IEEE Int. Workshop on Machine Learning for Signal Processing (MLSP)*, Sept. 2009, pp. 1–6.
- [20] O. Stegle, S. Fallert, D. MacKay, and S. Brage, "Gaussian process robust regression for noisy heart rate data," *IEEE Transactions on Biomedical Engineering*, vol. 55, no. 9, pp. 2143–2151, Sept. 2008.
- [21] S. Monzón, T. Trigano, D. Luengo, and A. Artés-Rodríguez, "Sparse spectral analysis of atrial fibrillation electrograms," in *IEEE Int. Workshop on Machine Learning for Signal Processing (MLSP)*, Sept. 2012, pp. 1–6.
- [22] D. Luengo, J. Vía, S. Monzón, T. Trigano, and A. Artés-Rodríguez, "Cross-Products LASSO," in *IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, May 2013, pp. 6118–6122.
- [23] D. Luengo, J. Vía, and T. Trigano, "Efficient iteratively reweighted lasso algorithm for cross-products penalized sparse solutions," in *28th European Signal Proc. Conf. (EUSIPCO)*. IEEE, 2021, pp. 2045–2049.
- [24] T. Trigano and D. Luengo, "Intracardiac ECG pulse localization using overlapping block sparse reconstruction," *Biomedical Signal Processing and Control*, vol. 79, p. 103921, 2023.
- [25] N. Mourad, "ECG denoising based on successive local filtering," *Biomedical Signal Processing and Control*, vol. 73, p. 103431, 2022.
- [26] M. U. Zahid, S. Kiranyaz, T. Ince, O. C. Devcioglu, M. E. Chowdhury, A. Khandakar, A. Tahir, and M. Gabbouj, "Robust R-peak detection in low-quality holter ECGs using 1D convolutional neural network," *IEEE Trans. on Biomedical Eng.*, vol. 69, no. 1, pp. 119–128, 2021.
- [27] A. Peimankar and S. Puthusserypady, "DENS-ECG: a deep learning approach for ECG signal delineation," *Expert Systems with Applications*, vol. 165, p. 113911, 2021.
- [28] C. Han, W. Que, S. Wang, J. Zhang, J. Zhao, and L. Shi, "QRS complexes and T waves localization in multi-lead ECG signals based on deep learning and electrophysiology knowledge," *Expert Systems with Applications*, vol. 199, p. 117187, 2022.

Method	P-wave peak localization	QRS localization	T-wave peak localization
Ours (best results)	0.988	0.997	0.956
Ours (worst results)	0.926	0.970	0.926
Laguna [3]	0.973	0.999	0.956
Martinez [12]	0.995	0.997	0.975
Singh [49]	0.960	N/A	0.91
Di Marco [50]	0.979	1	0.973
Sun [51]	0.998	1	0.967
Martinez [52]	0.978	1	0.978
Vazquez [53]	0.998	1	0.968
Vitek [13]	0.993	1	0.92
Hughes [54]	0.960	0.995	0.953

TABLE II

PEAK LOCALIZATION RESULTS WITH **PPV (60)** ATTAINED BY SOME STATE-OF-THE-ART METHODS (RESULTS TAKEN FROM TABLE 4 IN [9]).

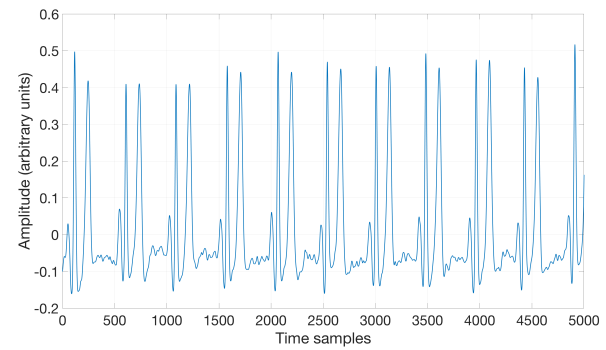
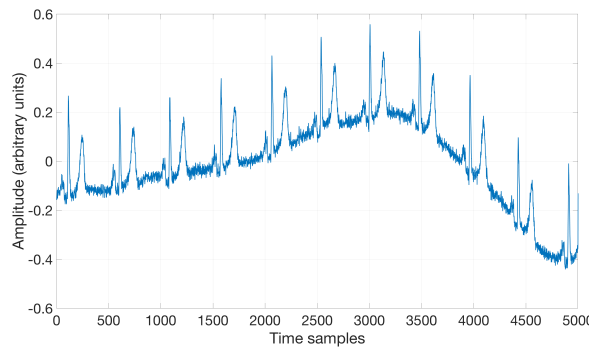


Fig. 23. Original signal (left) and denoised signal with AdapTrend (right), with arbitrary scales.

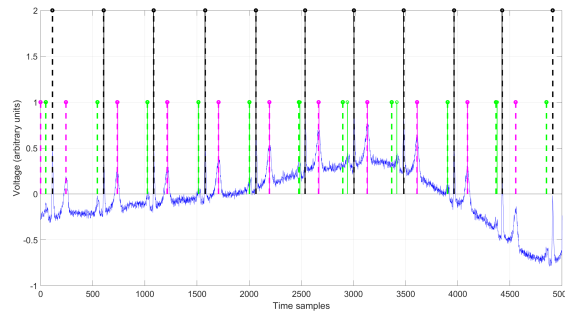


Fig. 24. Obtained peak localizations for the signal presented in Figure 23.

- [29] E. Fotiadou and R. Vullings, "Multi-channel fetal ECG denoising with deep convolutional neural networks," *Frontiers in Pediatrics*, vol. 8, p. 508, 2020.
- [30] F. P. Romero, D. C. Piñol, and C. R. Vázquez-Seisdedos, "DeepFilter: an ECG baseline wander removal filter using deep learning techniques," *Biomedical Signal Processing and Control*, vol. 70, p. 102992, 2021.
- [31] H. Li, G. Ditzler, J. Roveda, and A. Li, "DeScoD-ECG: deep score-based diffusion model for ECG baseline wander and noise removal," *IEEE Journal of Biomedical and Health Informatics*, 2023.
- [32] X. Wang, B. Chen, M. Zeng, Y. Wang, H. Liu, R. Liu, L. Tian, and X. Lu, "An ECG signal denoising method using conditional generative adversarial net," *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 7, pp. 2929–2940, 2022.
- [33] K. Nezamabadi, N. Sardaripour, B. Haghi, and M. Forouzanfar, "Un-supervised ECG analysis: A review," *IEEE Reviews in Biomedical Engineering*, 2022.
- [34] N. Faruk, A. Abdulkarim, I. Emmanuel, Y. Y. Folawiyi, K. S. Adewole, H. A. Mojeed, A. A. Oloyede, L. A. Olawoyin, I. A. Sikiru, M. Nehemiah *et al.*, "A comprehensive survey on low-cost ECG acquisition systems: Advances on design specifications, challenges and future direction," *Biocybernetics and Biomedical Engineering*, vol. 41, no. 2, pp. 474–502, 2021.
- [35] S. Kumar, D. Panigrahy, and P. K. Sahu, "Denoising of Electrocardiogram (ECG) signal by using empirical mode decomposition (EMD) with non-local mean (NLM) technique," *Biocybernetics and Biomedical Engineering*, vol. 38, no. 2, pp. 297–312, Jan. 2018.
- [36] A. Chacko and S. Ari, "Denoising of ECG signals using Empirical Mode Decomposition based technique," in *IEEE Int. Conf. on Advances in Eng., Science and Management (ICAESM)*, Mar. 2012, pp. 6–9.
- [37] B. Weng, M. Blanco-Velasco, and K. E. Barner, "ECG Denoising Based on the Empirical Mode Decomposition," in *2006 Int. Conf. of the IEEE Engineering in Medicine and Biology Soc.*, Aug. 2006, pp. 1–4.
- [38] X. Tan, X. Chen, X. Hu, R. Ren, B. Zhou, Z. Fang, and S. Xia, "EMD-Based Electrocardiogram Delineation for a Wearable Low-Power ECG Monitoring Device," *Canadian Journal of Electrical and Computer Engineering*, vol. 37, no. 4, pp. 212–221, 2014.
- [39] A. I. Kalyakulina, I. I. Yusipov, V. A. Moskalenko, A. V. Nikolskiy, K. A. Kosonogov, G. V. Osipov, N. Y. Zolotykh, and M. V. Ivanchenko,

- "LUIDB: A New Open-Access Validation Tool for Electrocardiogram Delineation Algorithms," *IEEE Access*, vol. 8, pp. 186 181–186 190, 2020.
- [40] F. Ferraty and P. Vieu, *Nonparametric Functional Data Analysis: Theory and Practice*, 1st ed. Springer Series in Statistics, 2006.
- [41] B. W. Silveanu and J. Ramsay, *Functional Data Analysis*, 2nd ed. Springer Series in Statistics, 2005.
- [42] G. Steidl, S. Didas, and J. Neumann, "Splines in Higher Order TV Regularization," *International Journal of Computer Vision*, vol. 70, no. 3, pp. 214–255, 2006.
- [43] S.-J. Kim, K. Koh, S. Boyd, and D. Gorinevsky, "L1 Trend Filter," *SIAM Review*, vol. 51, no. 2, pp. 339–360, 2009.
- [44] P. McSharry, G. Clifford, L. Tarassenko, and L. Smith, "A dynamical model for generating synthetic electrocardiogram signals," *IEEE Transactions on Biomedical Engineering*, vol. 50, no. 3, pp. 289–294, 2003.
- [45] G. D. Clifford, F. Azuaje, and P. McSharry, *Advanced Methods And Tools for ECG Data Analysis*, 1st ed. Artech House Pub., Sep. 2006.
- [46] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, *Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers*. Now Publishing, 2011.
- [47] A. Ramdas and R. J. Tibshirani, "Fast and Flexible ADMM Algorithms for Trend Filtering," *Journal of Computational and Graphical Statistics*, vol. 25, no. 3, pp. 839–858, Jul. 2016.
- [48] N. Pilia, C. Nagel, G. Lenis, S. Becker, O. Dössel, and A. Loewe, "ECGdeli - an open source ECG delineation toolbox for MATLAB," *SoftwareX*, vol. 13, p. 100639, 2021.
- [49] Y. N. Singh and P. Gupta, "ECG to Individual Identification," in *2008 IEEE Second International Conference on Biometrics: Theory, Applications and Systems*, Sep. 2008, pp. 1–8.
- [50] L. Y. Di Marco and L. Chiari, "A wavelet-based ECG delineation algorithm for 32-bit integer online processing," *Biomedical Engineering Online*, vol. 10, p. 23, Apr. 2011.
- [51] Y. Sun, K. L. Chan, and S. M. Krishnan, "Characteristic wave detection in ECG signal using morphological transform," *BMC Cardiovascular Disorders*, vol. 5, no. 1, p. 28, Sep. 2005.
- [52] A. Martinez, R. Alcaraz, and J. Rieta, "Application of the phasor transform for automatic delineation of single-lead ECG fiducial points," *Physiological Measurement*, vol. 31, no. 11, pp. 1467–1485, Nov. 2010.
- [53] C. R. Vazquez-Seisdedos, J. E. Neto, E. J. Maranon Reyes, A. Klautau, and R. C. Limao de Oliveira, "New approach for T-wave end detection on electrocardiogram: Performance in noisy conditions," *BioMedical Engineering OnLine*, vol. 10, no. 1, p. 77, Sep. 2011.
- [54] N. Hughes, L. Tarassenko, and S. J. Roberts, "Markov Models for Automated ECG Interval Analysis," in *Advances in Neural Information Processing Systems*, vol. 16. MIT Press, 2003.
- [55] M. A. Z. Fariha, R. Ikeura, S. Hayakawa, and S. Tsutsumi, "Analysis of Pan-Tompkins Algorithm Performance with Noisy ECG Signals," in *4th International Conference on Engineering Technology, Journal of Physics: Conference Series*, 2019, pp. 1–11.
- [56] D. Meltzer and D. Luengo, "Efficient clustering-based electrocardiographic biometric identification," *Expert Systems with Applications*, vol. 219, p. 119609, 2023.
- [57] C. D. Swerdlow, X. Zhang, R. Sawchuk, S. Greenhut, Y. Grinberg, Y. Liu, I. Crozier, D. O'Donnell, E. Kotschet, and H. Haqqani, "Design and preliminary results of sensing and detection for an extravascular implantable cardioverter-defibrillator," *Clinical Electrophysiology*, vol. 7, no. 11, pp. 1387–1399, 2021.